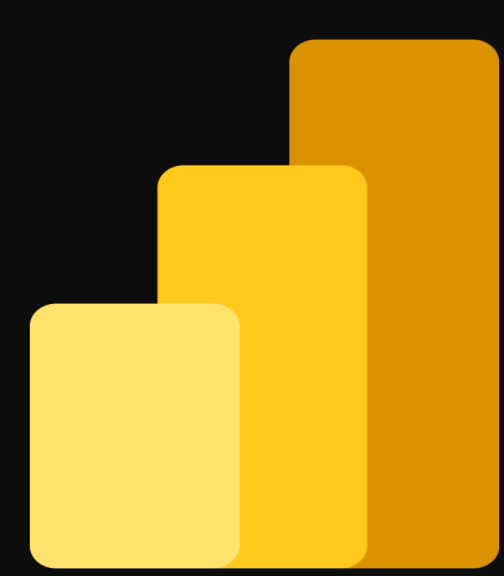


Data Modeling





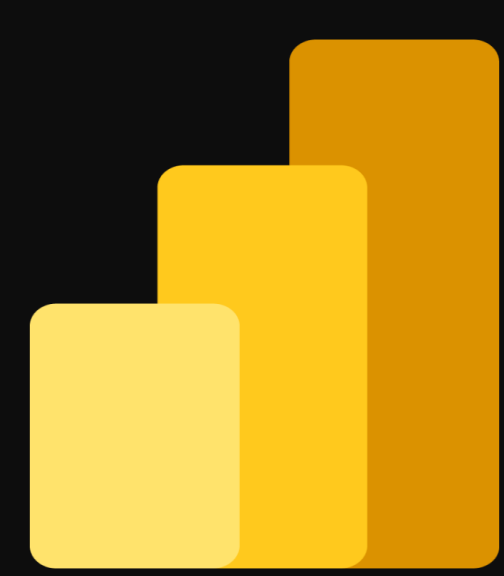
Defining Data Modeling

Definition:

- **Data modeling** is the process of **organizing data into logical tables** and **defining relationships** between them to support clear, consistent, and accurate analysis.
- It helps Power BI understand **how data from different sources connects**, ensuring that your visuals and calculations produce correct results.

Purpose of Data Modeling:

- To transform **raw data** into a **structured model** suitable for analysis.
- To define how tables **interact** and **filter** each other.
- To ensure reports remain **accurate**, **efficient**, and **scalable** as data grows.
- Just like a map helps you navigate a city, a data model helps Power BI navigate your data.



Key Concepts in Power BI Data Modeling

Fact Tables

- Contain **quantitative data** you want to analyze.
- Examples: Sales, transactions, revenue, order quantities.
- Usually have **numeric values** and foreign keys linking to dimension tables.

Dimension Tables

- Contain **descriptive or categorical data** that provides context for facts.
- Examples: Customers, Products, Regions, Time.
- Provide **filtering and grouping** capabilities in reports.



Key Concepts in Power BI Data Modeling

Definition:

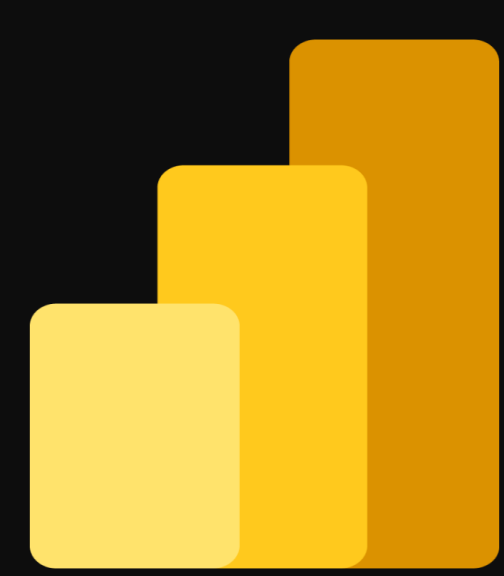
A relationship in Power BI connects two tables based on a common column, allowing data from multiple tables to interact in visuals and calculations.

Why Relationships Matter

- Ensures correct aggregation in reports.
- Helps create powerful insights without duplicating data.
- Allows you to build **fact tables** (transaction data) and **dimension tables** (reference data).
- Types of Relationships in Power BI

Cardinality defines the **number of unique records** that relate between two tables. It describes the **nature of the relationship**: how many rows in one table match rows in another.

Type	Description		Example
One-to-Many (1: *)	One record in Table A corresponds to multiple records in Table B.		Customers → Orders
Many-to-One (*:1)	Multiple records in Table A correspond to one record in Table B (inverse of above).		Orders → Customers
Many-to-Many (:)	Records in Table A correspond to multiple records in Table B and vice versa.		Products ↔ Suppliers
One-to-One (1:1)	Each record in Table A corresponds to exactly one record in Table B,		Each employee has exactly one email, and each email belongs to exactly one employee.



Key Concepts in Power BI Data Modeling

Schema:

The **structure of your data model** that defines how tables are organized and related.

- Determines **how data flows** between tables and **how calculations behave** in reports.

Star Schema:

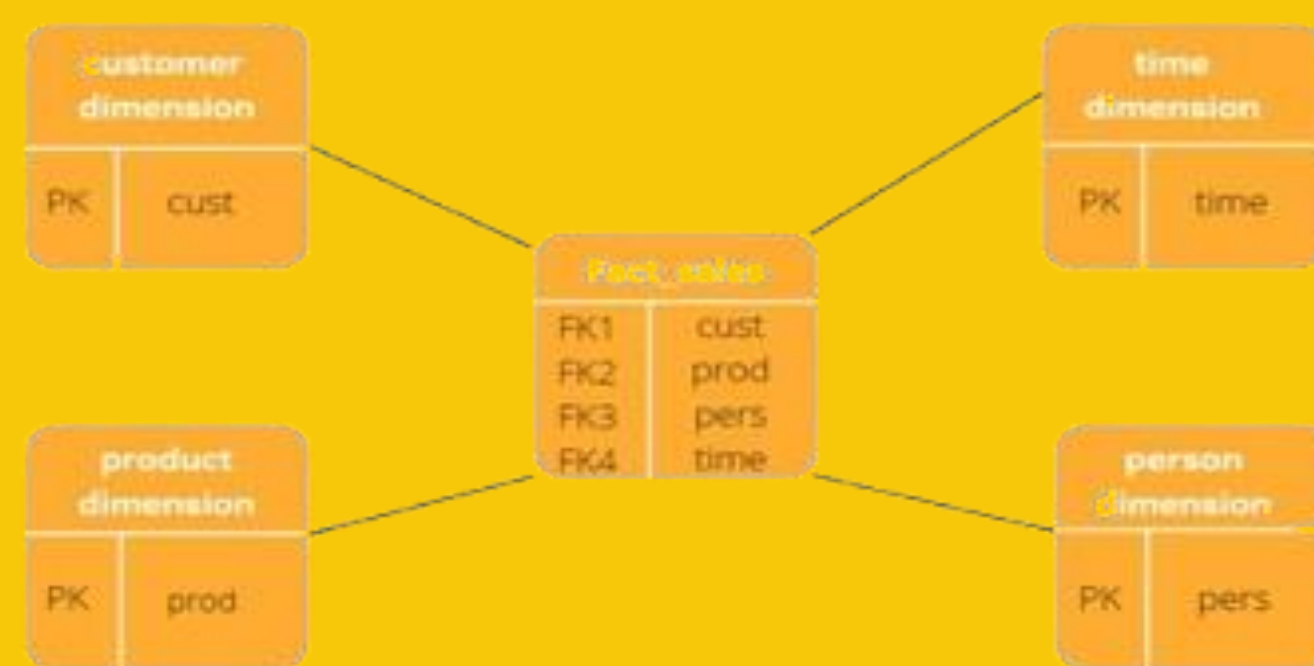
A schema where a **central fact table** connects directly to multiple **dimension tables**. Simple, fast for queries, and easy to understand.

Snowflake Schema:

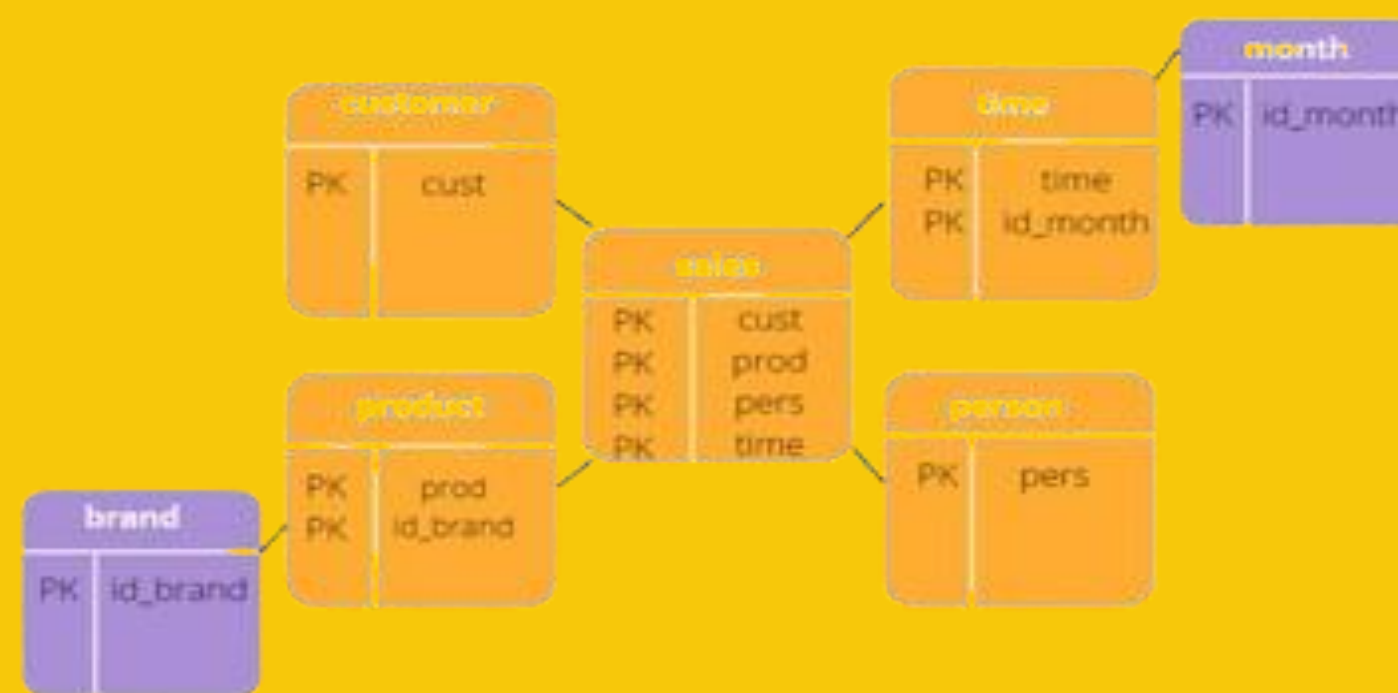
A schema where **dimension tables** are further normalized into related tables. More complex but saves storage and reduces data redundancy.

Star Schema Vs Snowflake schema

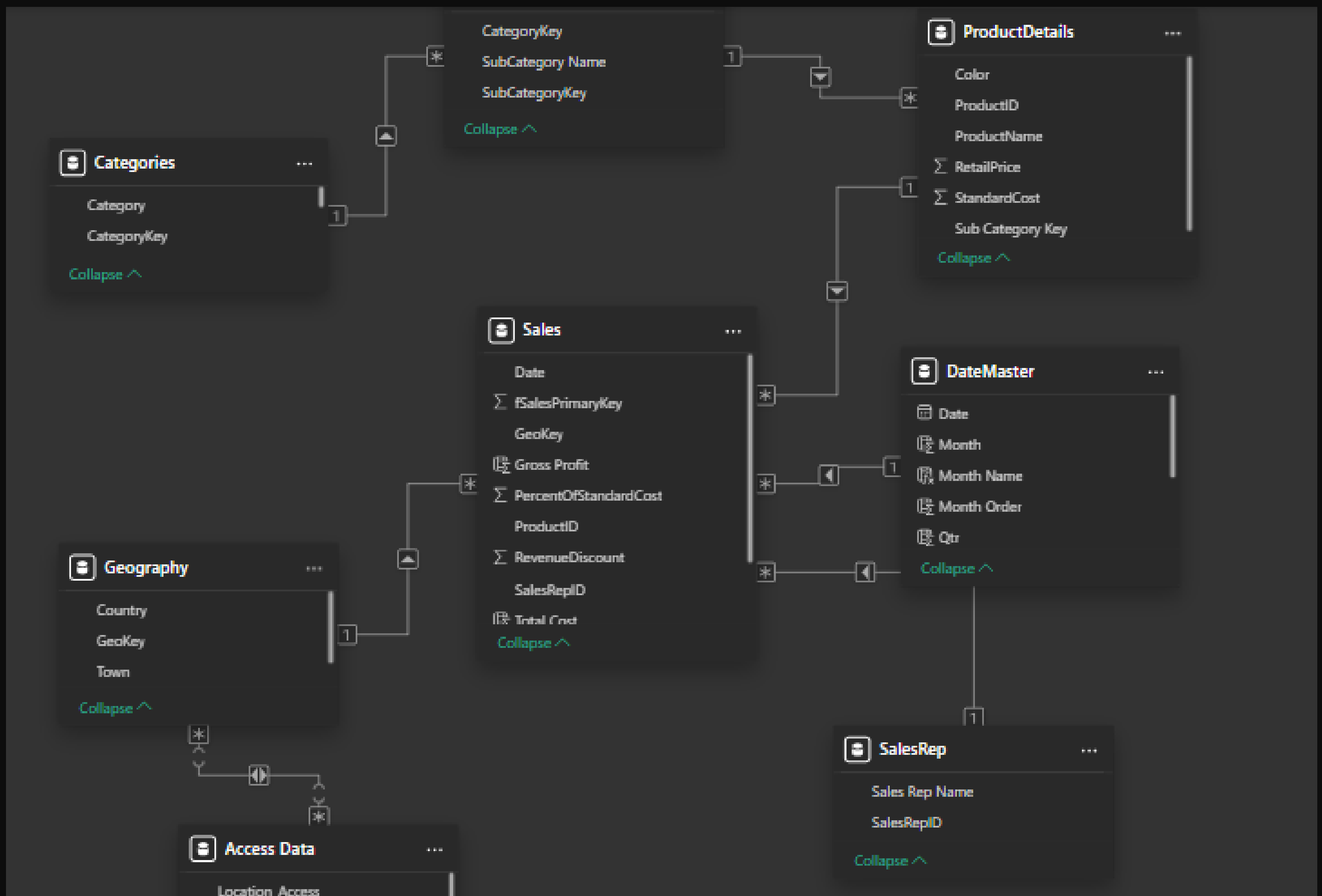
Star Schema



Snowflake Schema



The Power BI Data Model Example



This is the **data model** in Power BI — it visually represents how our data tables are organized and connected.

At the center, we have our *fact table* sales transactions, which stores the actual sales numbers.

Surrounding it are *dimension tables* describing products, customers, markets, and time.

Together, they form a **Star Schema**, which is ideal for analysis in Power BI.